

Arslan Shafique

ML Engineer | AI Engineer

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Portfolio: arslan-shafique.vercel.app

SUMMARY

Computer Science graduate and AI Engineer with hands-on experience building LLM/RAG applications, machine-learning systems, AI-powered IoT solutions and backend APIs. Experience with Python, FastAPI, PyTorch, LangChain, PostgreSQL, Docker, ESP32 and authentication and deployment, with strong focus on practical, reliable AI systems

EDUCATION

University of Central Punjab, Lahore

Bachelor of Computer Science,

Oct 2022 – July 2026

WORK EXPERIENCE

AI Engineer (Intern) - [Verify](#)

Techohub Systems (Lahore)

July 2026 – Sep 2026

- Build and extend Producix CoPilot and ERPNext-based solutions using LLMs, RAG, NLP, FastAPI, JWT authentication, PostgreSQL and Google APIs
- Develop containerized backend services and deployment configurations with Docker/ Docker Compose while integrating business data, authentication and AI-powered workflows.

Frontend Developer

Freelancing (Fiverr)

June 2020 – Oct 2022

- Delivered responsive and mobile-first websites for startup clients using HTML, CSS and Bootstrap.
- Translated client requirements into functional interfaces while managing, communication and project delivery.
- Applied reusable code, responsive design practices and performance optimisation across freelance projects..

PROJECTS

Smart Baby Band – Award Winning FYP

Final Year Project - [Source Link](#)

Oct 2025 – June 2026

ESP32-S3, Python, FastAPI, MQTT, Audio Processing

- Developed an AI-enabled infant monitoring prototype integrating health and motion sensors, audio-based baby cry detection and real-time data communication.
- Integrated the ESP32-S3 device with an MQTT communication layer using AWS IoT Core and FastAPI backend to transmit sensor readings and AI-generated events.
- Supported end-to-end integration between the embedded device, backend server and mobile application, including system testing and technical documentation.
- Collaborated with team members responsible for mobile and backend development to ensure reliable data flow across the complete system.

Financial Analysis Chatbot

Virtual Project – [Forage](#)

Sep 2026 – Sep 2026

Python, Pandas, Financial Data Analysis, Rule-Based Chatbot

- Calculated year-over-year growth trends and comparative financial performance across companies to identify changes in profitability, assets, liabilities, and cash flows.

- Built a rule-based chatbot prototype using Python functions and conditional logic to answer predefined financial queries directly from the analyzed dataset.
- Implemented dynamic company identification, financial metric retrieval, and year-over-year calculations to generate structured responses to user queries.
- Documented chatbot functionality, supported queries, test results, and limitations as part of the BCG GenAI Job Simulation on Forage.

Speech Emotion Recognition System

Semester Project - [Source Link](#)

May 2026 – May 2026

Python, MFCC, CNN, LSTM, RAVDESS

- Developed a five-class speech emotion recognition model using the RAVDESS audio dataset.
- Applied audio preprocessing, MFCC feature extraction and data augmentation before training a 2D CNN-LSTM architecture.
- Achieved 86.32% validation accuracy and a macro F1-score of 0.85.
- Evaluated model performance across emotion classes and documented the model-development workflow.

CERTIFICATION & COURSES

Generative AI Engineering with LLMs

- AI Agents Using RAG and LangChain
- Generative AI Advanced Fine-Tuning for LLMs
- Generative AI Engineering with Transformers & LLMs
- Generative AI Language Modeling with Transformers
- Gen AI Foundational Models for NLP & Language Understanding
- Generative AI and LLMs: Architecture and Data Preparation

[Verify](#)

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Python for Data Science and Machine Learning Bootcamp

[Verify](#)

Machine Learning, AI & Data Science

[Verify](#)

Foundations of LLMs and Deep Learning for Text Analysis

[Verify](#)

Basic Git and Github - essentials

[Verify](#)

Hive to ADVANCE Hive (Real-time usage): Hadoop Querying Tool on Udemy!

[Verify](#)

SKILLS

Programming: Python(intermediate), C++(intermediate), JS(basic), SQL(intermediate), HTML,CSS(expert)

AI & Machine Learning: LLMs, RAG, NLP, Machine Learning, Deep Learning, PyTorch, LSTM/RNN, OpenCV, Groq API

Generative AI & Agents: LangChain, Hugging Face, AI Agents, Vector Databases, Document Embeddings, Text Splitting, Gradio, FAISS, Prompt Engineering, Instruction Tuning, RLHF, PPO, DPO

Libraries and Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow PyTorch

Backend and Databases: FastAPI, JWT, PostgreSQL, MySQL, ERPNext

Tools: Git, GitHub, Jupyter Notebook

Core Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Database Management